# History of the science of genetics



**1** Unscramble the words in brackets to complete the information below.

Genetics is the study of how traits are passed from one (eeaiognrtn) _ _ _ _ _ _ _ _ _ _ to the next. Genetics also looks at why (glnvii) _ _ _ _ _ _ things of the same kind are so similar yet so unique.

People have known for hundreds of years that traits can be (deetnhirit) _ _ _ _ _ _ _ _ _. They observed how appearances were passed on from parent to child, and how illness ran through families, in both people and animals. But no one understood how inheritance really worked. The invention of the (oopirscemc) _ _ _ _ _ _ _ _ _ _ about 400 years ago allowed scientists to view tiny (lscle) _ _ _ _ _ for the first time. They soon discovered that living things were made of cells, and that cells divide to form new cells, which allows living things to (wrgo) _ _ _ _.

The first studies of inheritance were carried out by an Austrian monk named Mendel in 1866. He studied many (aaiihrterstcccs) _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ of the garden pea and cross-pollinated one at a time to discover that inheritance followed precise (eulsr) _ _ _ _ _.

By 1902, scientist Walter Sutton and biologist Theodore Boveri independently theorised that genes lie on (sooommchres) _ _ _ _ _ _ _ _ _ _ _ _. Their hypothesis was based on observations of how chromosomes behaved during (ssiiome) _ _ _ _ _ _ _, a type of cell division. Thomas Morgan went on to show that genes were located on chromosomes which themselves were located in the (suncule) _ _ _ _ _ _ _ of a cell.

In the 1940s, biologists George Beadle and Edward Tatum made major discoveries while studying the fungus Neurospora, a pink mould found on stale bread. They discovered that each gene produced one kind of (troipen) _ _ _ _ _ _ _. Soon after, biologist Oswald Avery demonstrated that genes are made up of deoxyribonucleic (caid) _ _ _ _, or DNA.

By the 1950s, scientists were striving to be the first to discover the three-dimensional structure of DNA. Biochemist Maurice Wilkins and scientist Rosalind Franklin worked together using X-ray diffraction crystallography. Franklin produced an image of DNA which showed that the DNA molecule is (eicllah) _ _ _ _ _ _ _with more than one (nachi) _ _ _ _ _. By 1953, biologists Francis Crick and James Watson had used Franklin's data to discover the three-dimensional (trusteruc) _ _ _ _ _ _ _ _ _ of DNA, which is now known as the double helix.

Gradually scientists began to unravel the mysteries of (snigeetc) _ _ _ _ _ _ _ _.
By the 1970s, experiments were conducted where foreign genes were inserted into a (caaribte) _ _ _ _ _ _ _ _ to successfully produce foreign protein. The development of modern research tools helped scientists learn more and at a much faster rate. For example, by 1986 Leroy Hood had developed a DNA (ceqseeun) _ _ _ _ _ _ _ _. This machine automatically reads the order of the building blocks of DNA.



By 1989, the Human (Gnomee) _ _ _ _ _ _ Project began. The aim of this project was to map the DNA sequence of humans. The project was completed in 2002.

Today, more powerful computers and new discoveries are teaching scientists a great deal about the genetic constructions of the human body and other living things. This new information is providing us with new opportunities to modify living things. However, it also means that people now face new and difficult (licetah) _ _ _ _ _ _ _ choices.

**2** List at least five ethical choices people now face due to technological developments in the area of genetics.

**3** Choose one ethical issue you listed in question 2. Research the issue and find out what the key points are. Decide what ***you*** think about the issue. Then write a persuasive text to communicate your opinion.

**4** Choose one scientist mentioned in the information above. Research the life of this scientist, and develop a presention explaining how this person has contributed to scientific knowledge.


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